WENHUI CUI

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Education

Northwestern Polytechnical University — **Xi'an, China**
Master, Software Engineering — Sep 2022 – Apr 2025

Southwest Jiaotong University — **Chengdu, China**
Bachelor, Network Engineering — Sep 2018 – Jun 2022

- National Encouragement Scholarship (2019, 2021, for top 10%)

Experience

General Software Engineer — **Jun 2025 – Present**
Huawei Technologies — *Shanghai, China*

- Designed and implemented benchmarking and performance analysis pipelines for the Rust standard library and Daft on ARM platforms
- Built automated regression tracking workflows and visualization tools for large-scale benchmark data analysis
- Conducted performance profiling and hotspot analysis using perf and assembly inspection to identify optimization opportunities
- Developed AI-assisted workflows for automated C/C++ to Rust migration and code quality analysis

Software Engineer Intern — **Jun 2023 – Sep 2023**
Kmerit — *Suzhou, China*

- Refactored a financial analytics platform using Apache Flink, reducing processing time for 100,000 assets from 60s to 20s through distributed stream processing
- Designed hierarchical aggregation pipelines for real-time financial indicators and streamed analytics results via Apache Kafka

Projects

Chinese Text Data Analysis & Input Scheme Generation | Python, Pandas, Jupyter Notebook

- Built data processing pipelines for large-scale Chinese linguistic resources, including character sets, phrase dictionaries, frequency statistics, and variant mappings
- Cleaned, normalized, and integrated structured and semi-structured text datasets to support downstream rule generation and analysis workflows
- Performed statistical analysis on character composition, phrase frequency, and text patterns to optimize input scheme generation rules

Rust Benchmarking Platform | Rust, postgresql, Vue

- Developed a benchmarking and analytics platform for collecting, visualizing, and tracking large-scale performance metrics
- Built automated benchmarking workflows and a Vue-based frontend dashboard for regression analysis and performance comparison
- Applied profiling and statistical analysis techniques to identify optimization opportunities across multiple workloads

Web Novel Reader | Rust, Flutter, Lua

- Developed a cross-platform content aggregation system with Rust backend and Flutter frontend, supporting concurrent task scheduling through an Actor-model architecture
- Built a sandboxed Lua-based parser engine for dynamic content extraction, supporting search, catalog parsing, pagination, and authenticated data retrieval workflows
- Designed local caching and incremental loading mechanisms to improve content synchronization efficiency and offline data access

Skills

Languages: Python, Rust, Java, SQL
Data & Infrastructure: Apache Flink, Apache Kafka, PostgreSQL, MySQL, Pandas
Analytics & Performance: Benchmarking, performance profiling, regression analysis, data visualization
Tools: Linux, Git, GitHub Actions